

Cube Roots of Perfect Cubes

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A **cube root** of a number is a value that, when multiplied by itself three times, gives the original number.

Perfect Cube

A **perfect cube** is a number that can be written as the cube of an integer.

$$27 = 3^3$$

Therefore, 27 is a perfect cube.

Cube Root Symbol

If $a^3 = b$, then a is the **cube root** of b .

$$\sqrt[3]{b} = a$$

For example:

$$\sqrt[3]{27} = 3$$

Finding Cube Roots

To determine the cube root of a perfect cube, identify the number that was multiplied by itself three times.

$$\sqrt[3]{64} = 4$$

because

$$4 \times 4 \times 4 = 64$$

Examples

Example 1

Find $\sqrt[3]{27}$.

$$\sqrt[3]{27} = 3$$

because

$$3 \times 3 \times 3 = 27$$

Example 2

Find $\sqrt[3]{125}$.

$$\sqrt[3]{125} = 5$$

because

$$5 \times 5 \times 5 = 125$$

Example 3

Find $\sqrt[3]{216}$.

$$\sqrt[3]{216} = 6$$

because

$$6 \times 6 \times 6 = 216$$

Practice Exercises

Practice 1

Find the cube root.

$$\sqrt[3]{8}$$

Practice 2

Find the cube root.

$$\sqrt[3]{64}$$

Practice 3

Find the cube root.

$$\sqrt[3]{216}$$

Practice 4

Find the cube root.

$$\sqrt[3]{343}$$

Practice 5

Find the cube root.

$$\sqrt[3]{729}$$

Practice 6

Find the cube root.

$$\sqrt[3]{1000}$$

Practice 7

Find the cube root.

$$\sqrt[3]{1728}$$

Practice 8

Find the cube root.

$$\sqrt[3]{27}$$

Practice 9

Find the cube root.

$$\sqrt[3]{125}$$

Practice 10

Find the cube root.

$$\sqrt[3]{512}$$

Activities

Activity 1

Determine the cube root.

$$\sqrt[3]{1}$$

Activity 2

Determine the cube root.

$$\sqrt[3]{125}$$

Activity 3

Determine the cube root.

$$\sqrt[3]{512}$$

Activity 4

Determine the cube root.

$$\sqrt[3]{1000}$$

Activity 5

Determine the cube root.

$$\sqrt[3]{1331}$$

Activity 6

Determine the cube root.

$$\sqrt[3]{2197}$$

Activity 7

Determine the cube root.

$$\sqrt[3]{2744}$$

Activity 8

Determine the cube root.

$$\sqrt[3]{8}$$

Activity 9

Determine the cube root.

$$\sqrt[3]{64}$$

Activity 10

Determine the cube root.

$$\sqrt[3]{343}$$

Thank you!